

# System Report: Enterprise Knowledge RAG Pipeline

## 1. Overview

This system implements a five-stage RAG pipeline for organizational documentation. The selected production configuration is `**`reranked_local`**` (cross-encoder reranking + Ollama ``qwen2.5:7b``) for fully on-premise, private inference.

## 2. Stage 1: Document Ingestion & Text Segmentation

### ### Extraction choices

| Format | Library                   | Rationale                                                                           |
|--------|---------------------------|-------------------------------------------------------------------------------------|
| PDF    | <code>`pdfplumber`</code> | Preserves reading order better than raw byte extraction; handles multi-page manuals |
| TXT/MD | Native read               | Zero-loss for pre-cleaned corpora                                                   |

### ### Noise removal

- Regex stripping of page numbers, "Confidential" footers, document ID/version/effective-date boilerplate
- Collapse of excessive whitespace and repeated layout artifacts
- Document type classification via filename + heading heuristics: ``standard_operating_procedure``, ``corporate_policy``, ``compliance_regulation``, ``technical_troubleshooting_log``

### ### Chunk boundaries

| Parameter  | Value                       | Justification                                                                  |
|------------|-----------------------------|--------------------------------------------------------------------------------|
| Chunk size | <code>**512 tokens**</code> | Balances retrieval precision vs context window; fits dense policy paragraphs   |
| Overlap    | <code>**64 tokens**</code>  | Prevents sentence splits at section boundaries (e.g., numbered policy clauses) |
| Min chunk  | <code>**100 chars**</code>  | Filters empty/noise fragments from PDF parsing                                 |

Segmentation first splits on paragraph boundaries and section headings (uppercase lines, numbered clauses), then applies token-based sliding windows within each section.

## 3. Stage 2: Embedding Generation & Indexing

#### ### Embedding model

**\*\*`sentence-transformers/all-MiniLM-L6-v2`\*\*** (384-dim, cosine similarity)

- Fast local inference, no API cost
- Strong baseline for short-to-medium enterprise text
- Normalized embeddings for stable cosine distance

#### ### Vector store

**\*\*ChromaDB\*\*** (persistent local index at `models/vector\_store/`)

| Setting         | Value                            |
|-----------------|----------------------------------|
| ----- -----     |                                  |
| Distance metric | Cosine (`hnsw:space: cosine`)    |
| Index type      | HNSW (Chroma default)            |
| Serialization   | Automatic via `PersistentClient` |

Metadata stored per chunk: `source\_file`, `doc\_type`, `section\_hint`, `token\_count`.

## 4. Stage 3: LLM Inference & Context Orchestration

#### ### Three operational paths

1. **\*\*`local\_llm`\*\*** - Top-k vector retrieval -> Ollama
2. **\*\*`reranked\_local`\*\*** - Top-8 vector search -> cross-encoder rerank to top-4 -> Ollama
3. **\*\*`extractive`\*\*** - Retrieval-only baseline (no LLM, for benchmark comparison)

#### ### Reranker

**\*\*`cross-encoder/ms-marco-MiniLM-L-6-v2`\*\*** re-scores (query, chunk) pairs to reduce noise from embedding-only retrieval.

#### ### Anti-hallucination guardrails

| Guardrail            | Implementation                                               |
|----------------------|--------------------------------------------------------------|
| ----- -----          |                                                              |
| Context-only answers | System prompt forbids external knowledge                     |
| Abstention           | Fixed phrase when no chunks pass similarity threshold (0.25) |
| Source citation      | Filenames required in LLM instructions                       |
| Context cap          | Max ~3000 tokens of retrieved text                           |

| Confidence score | Derived from top-k similarity (not LLM self-report) |

**\*\*Justification:\*\*** Groundedness is enforced at retrieval (threshold + rerank), generation (prompt), and response (source metadata + confidence from vector scores). Abstention prevents fabricated answers when retrieval fails.

## 5. Stage 4: Pipeline Evaluation

### ### Benchmark

12 question-answer pairs derived from the corpus (`data/eval\_questions.json`), covering all document types.

### ### Metrics

| Metric            | Definition            | Method                                          |
|-------------------|-----------------------|-------------------------------------------------|
| ----- ----- ----- |                       |                                                 |
| <b>**CR**</b>     | Context Relevance     | Max cosine sim(query, retrieved excerpts)       |
| <b>**F**</b>      | Faithfulness          | Mean max-sim(answer sentences, source excerpts) |
| <b>**AR**</b>     | Answer Relevance      | Cosine sim(query, answer)                       |
| <b>**L**</b>      | Latency               | End-to-end ms per query                         |
| <b>**QR**</b>     | Query Resolution Rate | % answers containing expected key facts         |

### ### Actual results (from `reports/metrics\_comparison.csv`)

| Pipeline                       | CR               | F                | AR               | QR               | L (ms)          |
|--------------------------------|------------------|------------------|------------------|------------------|-----------------|
| ----- ---- --- ---- ---- ----- |                  |                  |                  |                  |                 |
| local_llm                      | 0.624            | 0.692            | 0.687            | 0.769            | 6053            |
| <b>**reranked_local**</b>      | <b>**0.624**</b> | <b>**0.681**</b> | <b>**0.703**</b> | <b>**0.846**</b> | <b>**6100**</b> |
| extractive                     | 0.624            | 0.692            | 0.687            | 0.769            | 22              |

*Evaluated on 13 benchmark questions. Default model `qwen2.5:7b` (fits 8 GB RAM); use `llama3.1` if 16 GB+ available.*

### ### Selected configuration

**\*\*`reranked\_local`\*\*** - Reranking improves context precision; Ollama keeps all data on-premise with zero external API dependency.

## 6. Stage 5: API Deployment

- **\*\*Framework:\*\*** FastAPI with auto Swagger at `/docs`

- **Endpoint:** `POST /query` -> `{answer, confidence, sources}``
- **State serialization:** `models/state/pipeline_state.json`` stores active pipeline, prompt template, LLM/routing config
- **Index serialization:** ChromaDB persistent store in `models/vector_store/``

## 7. Error Review & Limitations

1. **PDF layout-heavy docs** - Scanned PDFs without OCR are not supported; text-based PDFs only.
2. **Faithfulness proxy** - Embedding-based F score approximates groundedness; human eval recommended for production.
3. **Ollama dependency** - Requires Ollama running locally; falls back to retrieval-only summary if unavailable.

## 8. Demo Checklist

- [x] Ingest unseen PDF: `backup_recovery_sop.pdf`` ingested (7 docs, 43 chunks)
- [x] Show vector distances in API response `sources[].similarity``
- [x] Present `reports/metrics_comparison.png``
- [x] Live `/query`` for situational prompts (VPN lockout, password policy, GDPR)

*Generated for E-Cell AI & Automation Task 2*